

Priority Queues: Binary Heaps

Data Structures

Data Structures and Algorithms

Sort Using Priority Queues

HeapSort($A[1 \dots n]$)

create an empty priority queue

for i from 1 to n :

 Insert($A[i]$)

for i from n downto 1:

$A[i] \leftarrow \text{ExtractMax}()$

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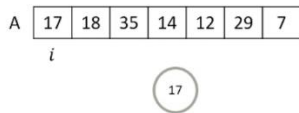
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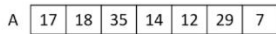
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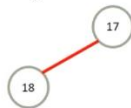
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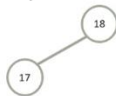
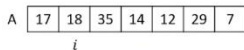
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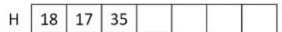
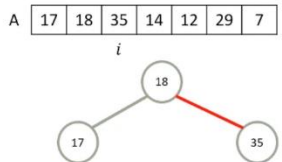
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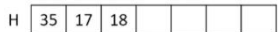
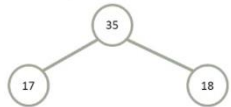
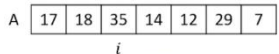
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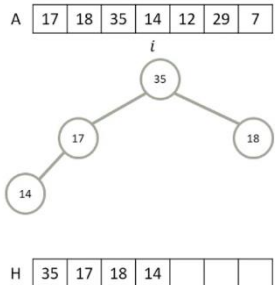
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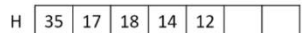
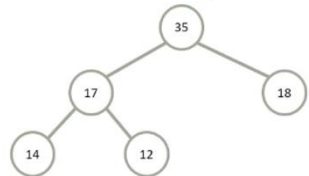
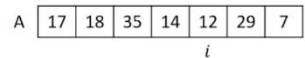
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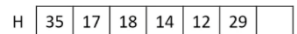
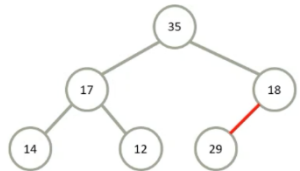
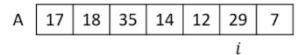
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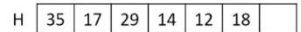
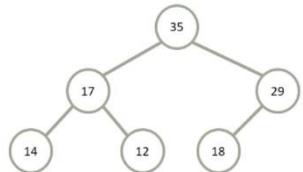
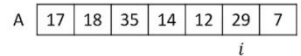
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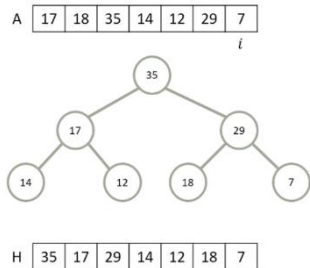
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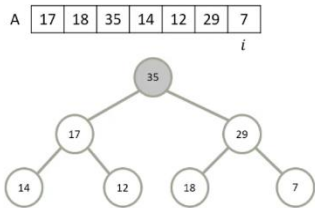
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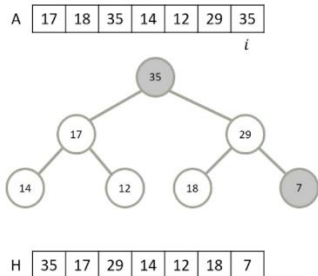
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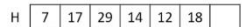
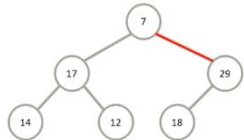
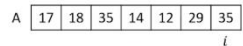
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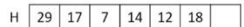
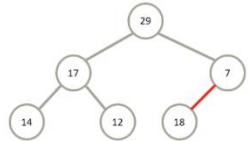
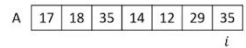
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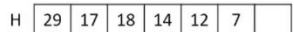
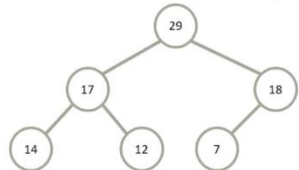
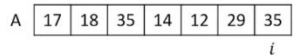
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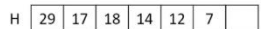
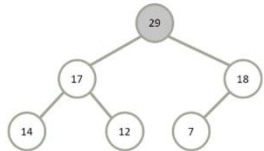
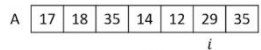
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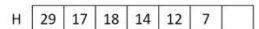
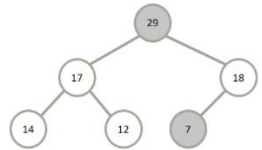
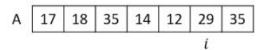
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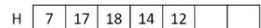
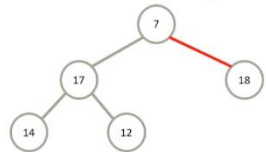
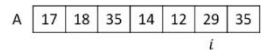
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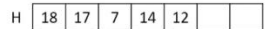
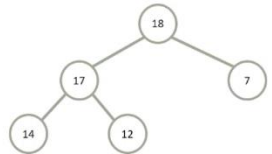
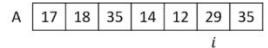
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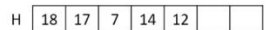
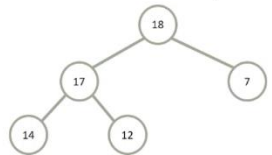
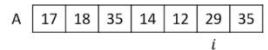
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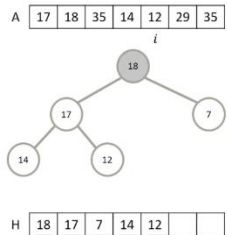
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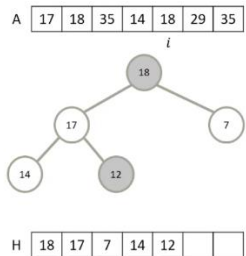
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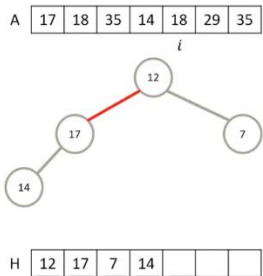
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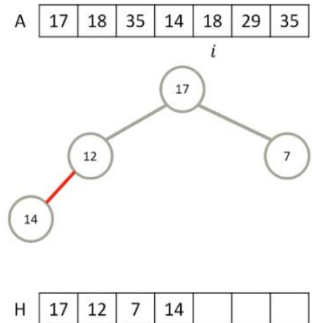
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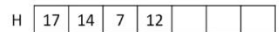
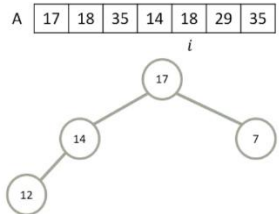
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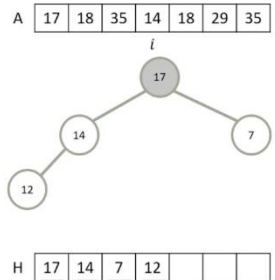
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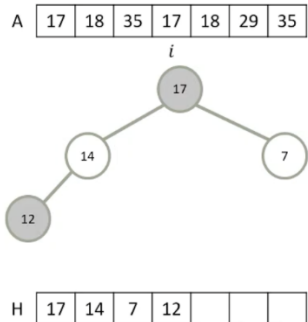
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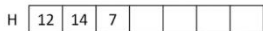
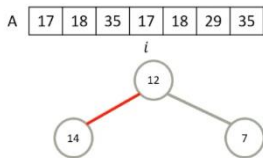
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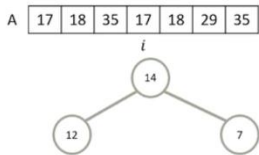
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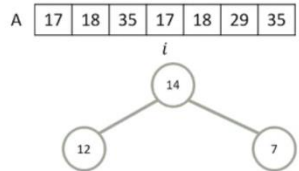
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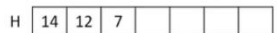
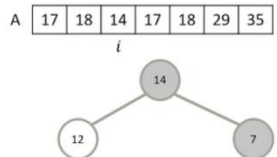
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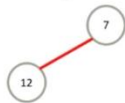
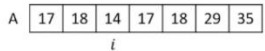
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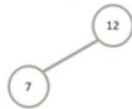
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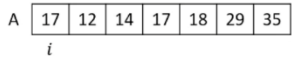
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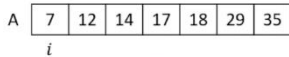
HeapSort($A[1 \dots n]$)

create an empty priority queue
for i from 1 to n :

 Insert($A[i]$)

for i from n downto 1 :

$A[i] \leftarrow \text{ExtractMax}()$



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A

7	12	14	17	18	29	35
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i

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